

Sungmin Kang

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Education

Texas A&M University

Ph.D in Computer Science (Advisor: *Prof. Zhengzhong Tu*)

College Station, TX, United States

Aug. 2026 - Present

University of Southern California (USC)

M.S. in Electrical Engineering (Advisor: *Prof. Salman Avestimehr*)

GPA: 4.00/4.00

Los Angeles, CA, United States

Aug. 2024 - May. 2026

MS Honors Fellow

Sogang University

B.S. in Electronic Engineering

GPA: 3.91/4.3

Seoul, Republic of Korea

Mar. 2018 - Feb. 2024

Magna Cum Laude

Research Interests

Multimodal Generation, World Models, Agentic AI Systems, Trustworthy LLMs, Interpretability, Federated Learning

Publications

- C5. **Uncertainty as Feature Gaps: Epistemic Uncertainty Quantification of LLMs in Contextual Question-Answering**
Yavuz Faruk Bakman, Sungmin Kang, Zhiqi Huang, Duygu Nur Yaldiz, Catarina G Belém, Chenyang Zhu, Anoop Kumar, Alfey Samuel, Daben Liu, Salman Avestimehr, Sai Praneeth Karimireddy
The Fourteenth International Conference on Learning Representations (ICLR 2026) [PDF]
- C4. **GEM: A Scale- and Distribution-Aware Sparse Fine-Tuning Framework for Effective Downstream Adaptation**
Sungmin Kang, Jisoo Kim, Salman Avestimehr, Sunwoo Lee
The 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026) [PDF]
- C3. **Layer-wise Update Aggregation with Recycling for Communication-Efficient Federated Learning**
Jisoo Kim, Sungmin Kang, Sunwoo Lee
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025) [PDF]
- C2. **TruthTorchLM: A Comprehensive Library for Predicting Truthfulness in LLM Outputs**
Duygu Nur Yaldiz*, Yavuz Faruk Bakman*, Sungmin Kang, Alperen Ozis, Hayrettin Eren Yildiz, Mitash Shah, Zhiqi Huang, Anoop Kumar, Alfey Samuel, Daben Liu, Sai Praneeth Karimireddy, Salman Avestimehr
The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025) [PDF] [Github]
- C1. **Reconsidering LLM Uncertainty Estimation Methods in the Wild**
Yavuz Faruk Bakman*, Duygu Nur Yaldiz*, Sungmin Kang, Tuo Zhang, Baturalp Buyukates, Salman Avestimehr, Sai Praneeth Karimireddy
63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025) [PDF]
- M1. **Uncertainty Quantification for Hallucination Detection in Large Language Models: Foundations, Methodology, and Future Directions**
Sungmin Kang, Yavuz Faruk Bakman, Duygu Nur Yaldiz, Baturalp Buyukates, Salman Avestimehr
IEEE BITS the Information Theory Magazine, 2025 [PDF]

Work Experiences

Research Intern, Argonne National Laboratory

Jun. 2026 - Aug. 2026

Host: *Dr. Kibaek Kim*, Mathematics and Computer Science Division

- **Federated Learning Simulator**: Built a simulator that faithfully reproduces realistic FL environments—from large-scale cross-device settings to high-performance computing—and reports system cost (compute, communication, queueing)
- **Agentic Paper-to-Code for FL**: Developed an LLM-agent pipeline that reads an FL research paper and automatically implements it on existing frameworks (APPFL, Flower, NVFlare), producing framework-compliant code verified by an automated rubric

Research Experiences

Graduate Research Assistant, University of Southern California

Oct. 2024 - Present

Advisor: *Prof. Salman Avestimehr*, Information Theory and Machine Learning (vITAL) Lab

- **LLM Uncertainty Quantification** [C1, C5]: Proposed a UQ method that measures a model's feature gap in its representation space using contrastive prompts; empirically evaluated existing UQ methods under realistic deployment conditions such as calibration, robustness to input variation, and ensembling
- **Open-Source & Survey** [C2, M1]: Built a truthfulness-evaluation library (25+ methods) and authored a survey that categorizes and organizes 35 representative UQ methods

Graduate Research Intern, Inha University

May. 2024 - Dec. 2025

with *Prof. Sunwoo Lee*, Large-Scale Machine Learning Systems Lab

- **Communication-Efficient Federated Learning** [C3]: Designed an FL algorithm that selects and recycles layer updates to reduce communication cost
- **Parameter-Efficient Fine-Tuning** [C4]: Proposed a sparse fine-tuning framework that uses the gradient-to-weight ratio to identify and tune only the important parameters for effective downstream adaptation

Undergraduate Research Assistant, Sogang University

Jul. 2023 - Apr. 2024

Advisor: *Prof. Hongseok Kim*, Networking for Intelligence Computing and Energy (NICE) Lab

- **Federated Learning Systems**: Built a real wireless FL system on 8 laptops with a multi-threaded server communicating over sockets [Github]
- **Distributed Optimization for FL**: Designed an FL algorithm that predicts client updates based on a distributed-optimization formulation for accelerated convergence

Undergraduate Research Intern, Seoul National University

Jan. 2023 - Apr. 2023

with *Prof. Taesup Moon*, Machine Intelligence and Data science (M.IN.D) Lab

- **Multimodal Medical Imaging**: MRI classification for predicting Alzheimer's disease progression via transfer learning, addressing limited data and class imbalance

Projects

TruthTorchLM: Open-Source Hallucination Detection Library, USC & Capital One

Oct. 2024 - May 2025

- Developed an open-source library with 25+ methods for evaluating LLM truthfulness [Github] [PDF]
- Provides a unified interface to calibrate, evaluate UQ methods, supporting long-form and contextual QA for both API and HuggingFace models

Honors and Awards

- MS Honors Fellowship, USC May 2026
- Outstanding Academic Achievement Award (one master's student department-wide), USC May 2026
- Student Travel Scholarship & Volunteer, AAAI-26 Jan. 2026
- Best Poster Award (out of 110+ teams), ECE 15th Annual Research Festival, USC Oct. 2025
- Best Poster Award (out of 50+ teams), Viterbi PhD Visit Day, USC Apr. 2025
- 3rd Prize (out of 50+ teams), Senior Thesis Project Contest, Sogang University Dec. 2023
- 3rd Prize (out of 40+ teams), Capstone Design Project Contest, Sogang University Jun. 2023
- Daesang Foundation Scholarship (Merit-based scholarship, 5M KRW/semester) Spring 2019 - Fall 2023

Teaching Experiences

- Grader, EE 503: Probability for Engineers by *Prof. Kosko Bart*, USC Fall 2025
- Teaching Assistant, EEE 4171: AI Communications by *Prof. Hongseok Kim*, Sogang University Spring 2024
- Teaching Assistant, COR 1010: AI Programming by *Prof. Naeun Jang*, Sogang University Summer 2023
- Education Volunteer Services for financially disadvantaged teenage students 2019, 2021

Academic Services

- *Invited Talk*: "Federated Learning Implementation through Socket Communication", Sogang University May 2024
- *Reviewer*: IEEE Transactions on Artificial Intelligence